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High Performance Processors and Systems

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Name
Last Name

Problem 1 (6 points)	
Problem 2 (6 points)	
Problem 3 (4 points)	
Question 1 (6 points)	
Question 2 (6 points)	
Question 3 (4 points)	
Total (32 points)	

To pass the exam (the following conditions are in AND):

- The overall score has to be equal or higher than 18
- At least 1 problem has to be correct (at least it has to be evaluated with 85% of the “available” points)
- At least 8 points from the problems
- At least 9 points from the questions

Problem 1

Design and describe (all the decisions have to be motivated) a 1-BHT and a 2-BHT able to execute the following assembly code.

(R0 is set to 100, R1 is set to 0)

```
LOOP:    LD      F1  0    R0
         ADDD    F2  F1   F1
         ADDI    R1  R1   300
LOOP2:   MULTD   F2  F2   F1
         SUBI    R1  R1   1
         BNEZ    R1  LOOP2
         SUBI    R0  R0   2
         BNEZ    R0  LOOP
```

The obtained result, in terms of miss predictions, is inline with theoretical characteristics of the two predictors? Please effectively support your answer.

Answer

Problem 2

Assume that the following code is executed on a CPU with SCOREBOARD and with the following units:

- 1 LOAD/STORE unit with Latency= 1
- 2 MULT units with latency = 10
- 1 DIVIDE unit with latency = 30
- 1 ADD/SUBD unit with latency 2

	Issue	Read Op	Exec Co.	Write R.
LD F6 32+ R2	1	2	3	4
LD F2 45+ R3	5	6	7	8
MULTD F0 F4 F2	6	9	19	20
ADD F8 F2 F6	7	9	10	11
DIVD F12 F8 F6	8	21	51	52
SUBD F8 F6 F2	9	13	14	15

Is the Pipeline schema correct?

If no, explain in details all the problems.

Problem 3

Consider two CPUs: CPU1 e CPU2. CPU1 has clock cycle of 2 ns while CPU2 has an operating frequency of 700MHz. Given the following frequencies of occurrence of the instructions for the two CPUs

Operation type	Frequency	CPU1	CPU2
A	0.3	2	2
B	0.1	3	3
C	0.2	4	3
D	0.3	2	2
E	0.1	4	3

a) Compute the average CPI for CPU1 and CPU2

b) Which is the slowest CPU?

c) Assume that CPU2 is modified in such a way that:

- All instructions require 8 clock cycles;
- Memory stalls are not considered;
- The miss penalty is 7 clock cycles

Assuming a miss rate of 23% and assuming 3 memory references on average for each instruction determine the impact on performance of the cache with respect to an ideal cache?

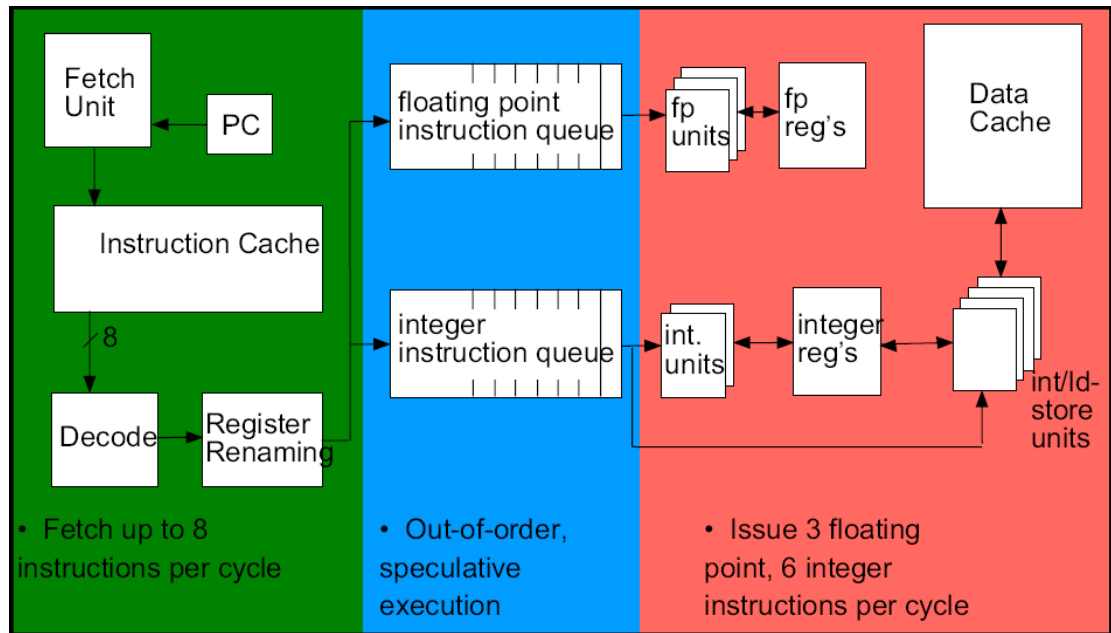
Question 1

For each of these techniques that improve cache performance, compile the first three columns with a +,- or nothing if they improve, worsen or do not impact on hit time, miss penalty and miss rate, and provide a short motivation for your answer. If you need more space for the motivation, please write at the end of the table.

Technique	Hit Time	Miss penalty	Miss rate	Motivation
Small and simple caches				
Trace caches				
Nonblocking caches				
Victim Caches				
Compiler techniques to reduce cache misses				
L2 cache				

Question 2

1. Given a processor oriented to Instruction Level Parallelism such as the one in figure, can it be exploited for thread level parallelism? Motivate your answer.
2. Which kind of Thread level parallelism is best suited and which modifications are necessary to this architecture?
3. How this can improve performance? Show the performance formula and explain which terms of the performance formula are concerned with thread level parallelism. Motivate your answer.



Question 3

Define the concept of precise exception and explain how to obtain it in superscalar processors with dynamic scheduling

